

Employment Continuity Around Childbirth in Japan's Dual Labour Market

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Japan: generous leave, large interruptions, a divided labour market

Japan combines three features that are hard to reconcile:

- **Comparatively generous formal leave entitlements:** childcare leave can extend until the child turns two when, for example, nursery admission is unavailable (MHLW; OECD PF2.1).
- **Low maternal employment when children are very young:** maternal employment is considerably lower when the youngest child is aged 0–2 than when children are older (OECD Family Database, LMF1.2).
- **A strongly segmented labour market:** a durable divide between regular (*seishain*) and non-regular (*hi-seiki*) employment, particularly important for women (OECD, 2024).

When women stop working around childbirth, do they still have a job to return to?

The position at first birth has a history

schooling, field of study and expectations about work and family



first employer and occupation



regular or non-regular employment track



marriage and anticipated care



first birth



still working, on leave, or job lost

By the time of a first birth, a woman may have spent a decade moving through these stages. Some steps are choices and some are constraints; many are choices made in anticipation of constraints.

The paper enters this sequence at its last step: the employment relationship a woman holds just before the birth-report wave, and whether it survives.

Leave requires an employment relationship that can survive

Regular employment (seishain) is organised around continuity: open-ended contracts, internal progression and an expected return after an absence.

Non-regular employment (hi-seiki) can offer more control over working time, and often comes with weaker continuity and career progression. It includes part-time, fixed-term and dispatch positions.

Family policy runs through that relationship. For much of the sample period, fixed-term workers faced additional tenure and contract-continuation conditions for childcare leave. Income replacement also operated through employment insurance and required a qualifying contribution history (MHLW). Formal rights therefore depended on an employment relationship capable of surviving the absence.

Paid work and care remain strongly gendered. Long-hours regular employment and expectations about maternal care leave households little room to redistribute the time cost of a child (JILPT, 2019).

Women make up roughly two-thirds of Japan's non-regular workforce (MHLW, 2025).

How women describe non-regular work

A 2026 JILPT survey of women workers, separate from the panel used below. Among respondents working outside regular employment, the most common reasons given were:

29.9%

“I did not think I could work in the way regular employees do.”

25.2%

Childcare or eldercare responsibilities.

14.0%

Wanted regular employment but could not obtain it.

Reading

Observed employment choices reflect the jobs women can obtain, the way those jobs are organised and the care they expect to provide.

Source: JILPT, 2026.

Two margins: working now, and keeping the job

Of the sequence on slide 2, the data observe the last step: the job a woman holds before the birth-report wave, and what becomes of it. I separate two outcomes that are usually folded into one “employment” measure:

- A. Current work** - is the woman doing paid work in the survey reference month?
- B. Employment attachment** - is she working *or* temporarily absent while retaining a job?

A woman on childcare leave is recorded as not currently working but employment-attached.

Question. Does pre-birth contract status predict whether an interruption is a *temporary absence* or a *loss of the job*?

Preview. Current work is much lower in the birth-report wave for both regular and non-regular women. The regular/non-regular divide appears in whether that interruption happens inside an ongoing employment relationship or ends it.

Related literature: Kleven, Landaís and Søgaaard (2019); Kleven et al. (2019); Goldin (2014); Yamaguchi (2019); Kikuchi (2026).

Data and timing

Data. Harmonised KHPS/JHPS household panels, 2004–2022. I follow 631 women around a birth reported in the annual survey (appendix A).

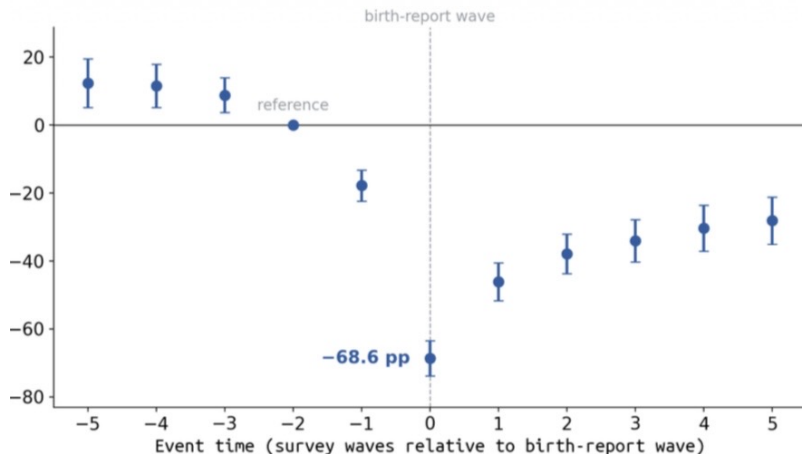
Event time. $t = 0$ is the wave in which the household reports a birth since the previous annual survey.

$$t = -2 \rightarrow t = -1 \rightarrow \underbrace{t = 0}_{\text{birth-report wave}} \rightarrow t = +1 \rightarrow \dots$$

The woman may already have been pregnant at $t = -1$, or the pregnancy and birth may both fall between $t = -1$ and $t = 0$, so $t = -2$ is the pre-birth reference. Event time is counted in survey waves, and the birth is located only to within a wave.

Contract status. Regular or non-regular, as classified by the type of position at the workplace, not by hours worked (Statistics Bureau of Japan, 2022).

Current work around the birth-report wave

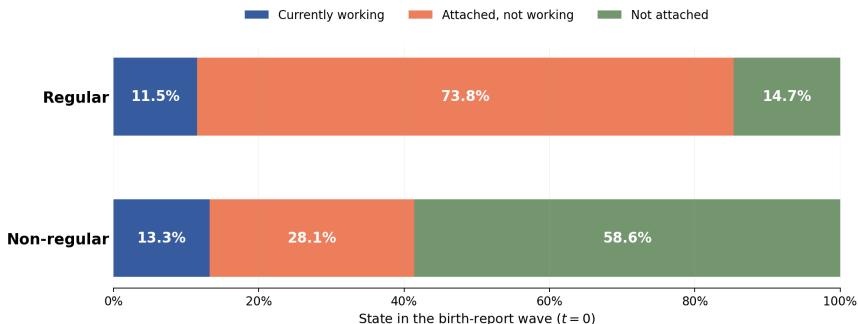


Current work is lowest in the birth-report wave and already lower at $t = -1$.

Coefficients relative to $t = -2$; calendar-year fixed effects; episode-clustered 95% CIs;

Same interruption, different employment continuity

All women below are employment-attached at $t = -1$.

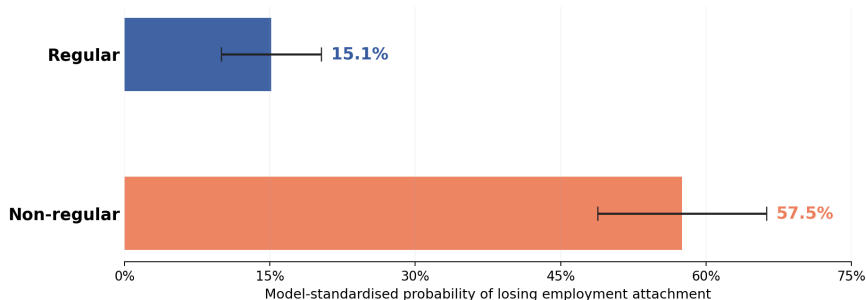


In the full $t = 0$ cohort, **194 of 205 attached non-workers (94.6%)** report **childcare leave** as the reason for their absence.

Raw shares among 319 classifiable wage workers who were employment-attached at $t = -1$ and observed at $t = 0$. Contract type at $t = -1$. The reason item names childcare leave; statutory maternity leave is not a separate category.

Contract status predicts attachment loss

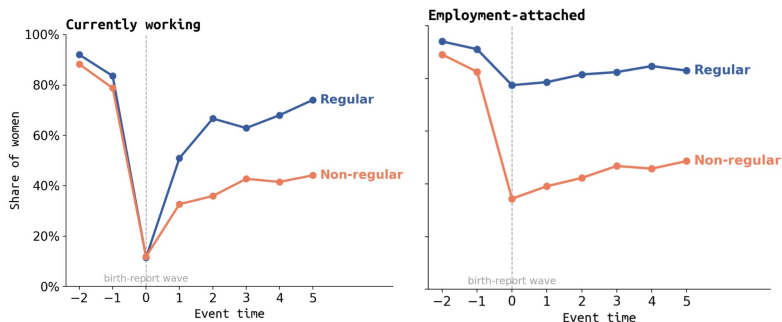
Classifiable wage workers attached at $t = -1$ and observed at $t = 0$ ($N = 319$).



Difference: 42.4 percentage points (95% CI: 32.2 to 52.6)

Model-standardised probabilities from a logit controlling for small-firm status and missing firm size; HC1 95% CIs. The adjusted odds ratio for non-regular status is 7.80 [4.54, 13.40].

Regular workers retain attachment; non-regular workers often do not



Raw shares over event time by fixed pre-birth contract group (last valid wage-job classification in $t = -5, \dots, -1$). Current work is lowest at $t = 0$ in both groups, with faster recovery among regular workers. Employment attachment remains around 77–85% for regular workers after $t = 0$; for non-regular workers it falls to 34% and reaches only 49% by $t = +5$. Fewer women contribute at longer horizons (appendix L).

The inequality lies in what remains when work stops

At the birth-report wave, few women in either group are currently working (11.5% of regular and 13.3% of non-regular workers). For regular workers the interruption usually happens inside an ongoing employment relationship: 73.8% are absent and still hold their job. For non-regular workers it far more often coincides with the loss of that relationship: 58.6% no longer hold one. The divide is of similar size within professional, clerical and sales-and-service occupations (appendix P).

The result is consistent with a leave system whose usefulness depends on the thing that differs between the two groups: whether the employment relationship is expected to continue (slide 3). Two women encounter the same family transition and are left in different employment states. One has a job to return to; the other would have to find a new one in order to return.

The life-course context (slide 2) explains why contract status is more than a label. The result here is narrower and sharper: when first birth interrupts work, the employment relationship a woman brings to that moment strongly predicts whether the interruption becomes leave or exit.

A common transition, unequal employment biographies

First birth interrupts active work in both contract groups. The difference is whether a job remains.

For regular workers, the interruption usually happens inside an employment relationship that continues. For many non-regular workers, it coincides with the end of that relationship, so a return to work means finding a new employer.

The employment position a woman brings to first birth reflects a longer history of schooling, occupations, employers and anticipated family roles. This paper shows the moment at which those earlier differences become consequential.

A generous leave entitlement cannot deliver equal continuity when the employment relationships that carry it are unequal.

Thank you

Thank you.
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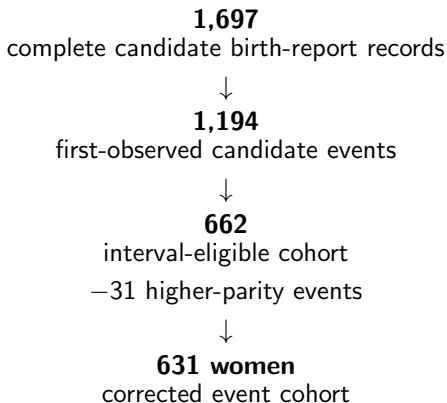
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Institutional and descriptive sources

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A. Sample construction



4,923	4,480	627
person-waves	determinate observations	focal-woman episodes

B. Outcome definitions and questionnaire routing

- **Current work:** questionnaire-reported paid work in the reference month; contradictory responses remain missing. Temporary absence while retaining a job does *not* count as currently working.
- **Employment attachment:** current work or job-attached temporary absence. Attached non-workers are labelled “employment-attached”; reported reasons for absence are on slide 8 and appendix O.
- **Weekly hours:** the job block is asked of women who worked and women temporarily absent while retaining a job. For the latter, reported hours describe the job’s usual hours, not realised hours in the reference month.

Routing diagnostic. Of 4,923 person-waves: 49.9% observed positive hours, 38.9% structural skips, 3.8% refusal/don’t-know, 7.0% focal-woman role unavailable, 0.5% contradictory. All 2,425 determinate valid hours reports are positive. Women in job-attached absence are also routed to the hours block (appendix F).

C. Event-study specification

$$Y_{it} = \sum_{k \neq -2} \beta_k \mathbf{1}\{\text{event time}_{it} = k\} + \gamma_t + \varepsilon_{it}$$

- Y_{it} is current work or employment attachment; the hours specification (appendix F) uses women who report current work.
- $t = -2$ is the presentation reference period.
- γ_t are calendar-year fixed effects.
- Pooled OLS; the coefficients are sample-average event-time trajectories in this panel.
- SEs clustered by focal-woman episode; robust and household-clustered SEs in appendix E.

D. Estimates with $t = -1$ as reference

Outcome (vs $t = -1$)	$t = 0$	$t = +1$	$t = +3$	$t = +5$
Current work	-0.508	-0.283	-0.163	-0.104
Employment attachment	-0.218	-0.204	-0.133	-0.133

Current work already falls at $t = -1$, so measured against $t = -1$ the $t = 0$ drop understates the full decline. The main slides use $t = -2$ as reference to show the decline from before it begins.

E. Inference comparison at $t = 0$ (reference $t = -2$)

Outcome	Coefficient	Robust SE	Clustered SE
Current work	-0.686	0.026	0.026
Employment attachment	-0.371	0.030	0.026

Household-clustered and episode-clustered SEs coincide (627 episodes); the three methods give the same inference.

F. Weekly hours by reference-month status

The hours question is asked of women who worked in the reference month and of women temporarily absent while retaining a job; for the latter, reported hours are the usual hours of the job they hold.

Coefficient vs $t = -2$	$t = -1$	$t = 0$	$t = +1$	$t = +3$	$t = +5$
All valid hours reports	-0.52	+1.05	-5.17	-5.84	-9.30
Currently working only	-1.01	-8.08	-7.69	-6.27	-9.35

Calendar-year fixed effects; episode-clustered SEs. At $t = 0$, 180 of 241 valid hours reporters are in job-attached absence and report 38.8 usual hours on average; the 61 women currently working report 27.2. The currently-working row is the labour-supply series.

G. What do the $t = 0$ hours reports contain?

Reference-month status	Valid hours reports	Mean reported hours
Currently working	61	27.2
Job-attached absence	180	38.8

Women in job-attached absence are routed through the hours block, so their reported hours describe the job they hold. Current work is measured from the reference-month status item, and hours are read among current workers.

Current work is therefore measured from the direct reference-month work-status item. Weekly hours are interpreted only among women who report that they are currently working.

H. Attachment-loss model

Risk set. Wage workers employment-attached at $t = -1$, with regular or non-regular contract status observed at $t = -1$ and a determinate $t = 0$ outcome.

Quantity	Estimate
Risk set N (attachment-loss events)	319 (103)
Raw loss: regular	14.7% (28/191)
Raw loss: non-regular	58.6% (75/128)
Adjusted OR, non-regular [95% CI]	7.80 [4.54, 13.40]
Predicted loss: regular [95% CI]	15.1% [10.0, 20.3]
Predicted loss: non-regular [95% CI]	57.5% [48.8, 66.2]
Adjusted difference [95% CI]	42.4 pp [32.2, 52.6]

Logit controlling for small-firm status and a missing-firm-size indicator. The full $N = 357$ specification with a missing indicator gives 42.7 pp and OR 7.87.

I. What this design can and cannot identify

- It **can** show the timing of work interruption and attachment loss around the birth-report wave, and where attachment loss is concentrated across pre-birth contract positions.
- It **can** separate temporary, job-attached absence from loss of the employment relationship.
- It **cannot** make contract type exogenous: selection into regular employment can reflect anticipated continuity.
- It **cannot** estimate the causal effect of a specific leave reform, and event time is survey-wave time rather than time from delivery.
- “Leave or exit” is conceptual shorthand: the data identify job-attached temporary absence; childcare leave is one identified reason, and statutory maternity leave is not separately identified.

J. Sample characteristics

Characteristic	Mean / share	SD / valid <i>N</i>
Mother age around focal birth	31.90 years	SD 4.32; <i>N</i> = 630
Partner age	33.78 years	SD 5.71; <i>N</i> = 631
Partner – mother age gap	1.89 years	SD 3.96; <i>N</i> = 630
Mother: university or higher	24.3%	valid <i>N</i> = 255
Partner: university or higher	48.4%	valid <i>N</i> = 399

“Partner” is the spouse recorded in the household at the focal birth.

Education is the latest completed attainment observed by the birth-report wave; valid *N* shown.

K. Education measurement

The education items record whether a respondent is **currently attending**, has **graduated** or has **dropped out**, so a reported level counts as attainment once completion is confirmed. Two measures follow from this.

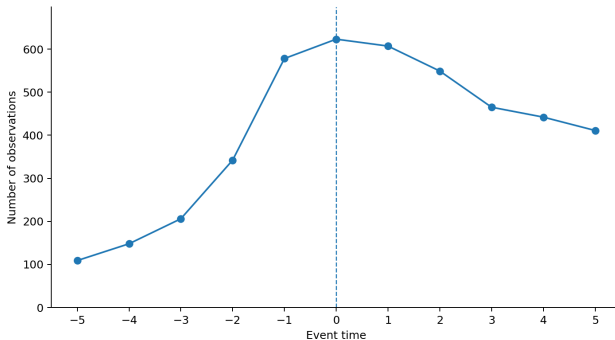
Primary / birth-wave measure

- + temporally appropriate (observed no later than the birth-report wave);
- + requires confirmed completed attainment;
- incomplete coverage: mothers $N = 255$, partners $N = 399$.

Full-history measure

- + much better coverage;
- may include education completed after childbirth;
- spouse-history identity can be ambiguous when spouse signatures change.

L. Event-time support in the unbalanced panel



Far leads require earlier panel entry and far lags continued observation, so fewer women contribute at longer horizons. The $t = 0$ contract comparison is within-wave, and the risk models compare the same women from $t = -1$ to $t = 0$. Support is identical for current work and employment attachment.

M. Balanced-panel contract check ($t = -2$ to $+1$)

The balanced sample contains the same 328 women with determinate current-work and attachment outcomes at every wave from $t = -2$ through $t = +1$. The contract comparison covers the 257 women with a classifiable pre-birth wage job (141 regular; 116 non-regular).

Outcome at $t = 0$	Sample	Regular	Non-regular
Currently working	Unbalanced headline	11.5%	11.9%
	Balanced panel	9.9%	10.3%
Employment-attached	Unbalanced headline	77.4%	34.4%
	Balanced panel	76.6%	33.6%

The attachment gap is 43.0 percentage points in both samples.

Contract group as on slide 10.

N. The contract divide persists after $t = 0$

	Pre-birth contract	$t = +1$	$t = +3$	$t = +5$
Currently working	Regular	51.0%	62.9%	74.1%
	Non-regular	32.7%	42.7%	44.1%
Still has a job	Regular	78.6%	82.4%	83.0%
	Non-regular	39.1%	46.8%	48.6%

- Regular workers return to current work while retaining a high level of job continuity: around 79–83% still have a job at each post-birth horizon.
- Non-regular workers are much less likely to retain a job: 34% do so in the birth-report wave, rising to about 49% by $t = +5$.

“Still has a job” includes women who are currently working and women temporarily absent while retaining an employment relationship. Contract group is the last valid wage-job classification in $t = -5, \dots, -1$ and is held fixed thereafter. Raw shares are descriptive and composition-sensitive in the unbalanced panel.

O. What is the attached-but-not-working state?

Determinate current-work status at $t = 0$	623
Employment-attached but not currently working	205
Explicit reported childcare leave	194
Named non-childcare reason	8
No recorded usable reason	3

Share of attached non-workers reporting childcare leave: **194/205 = 94.6%** of all cases; among rows with a recorded reason, **194/202 = 96.0%** report childcare leave. Of the 8 named non-childcare reasons, 7 are “other” and 1 is own health.

By fixed pre-birth contract group among attached non-workers: Regular **134/143 = 93.7%**; Non-regular **35/36 = 97.2%**. All 8 named non-childcare reasons occur in the regular group.

The reason item is asked only of respondents temporarily absent while retaining a job. “For childcare leave” is a named response category; statutory maternity leave is not separately identified. Contract groups here use the fixed pre-birth classification (slide 10), so counts differ from the $t = -1$ risk set (slide 9).

P. The contract divide appears within occupations

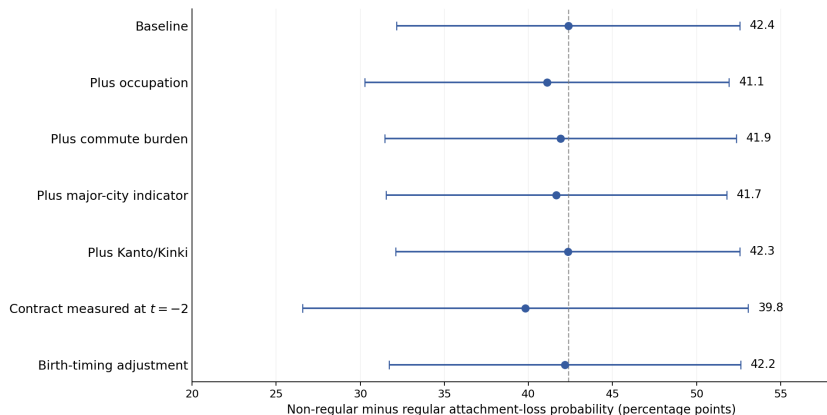
Displayed below are the three well-supported broad occupation groups among classifiable wage workers in the $t = -1$ attachment-risk set.

Broad occupation	Regular		Non-regular		Gap
	N	Loss	N	Loss	
Professional / managerial / IT	78	9.0%	27	59.3%	50.3 pp
Clerical	74	18.9%	41	58.5%	39.6 pp
Sales and services	31	19.4%	49	59.2%	39.8 pp

Non-regular attachment loss is about 59% in every well-supported occupational group.

Raw loss between $t = -1$ and $t = 0$. Adding broad occupation to the adjusted model moves the regular/non-regular probability difference from 42.4 to 41.1 pp. Occupation is itself a pre-birth sorting outcome, so it enters as a robustness check.

Q. The attachment gap is stable across specifications



Model-standardised non-regular minus regular attachment-loss probability, with HC1 95% CIs. The first five models use the same $N = 319$ classifiable wage-worker risk set; the $t = -2$ contract check uses $N = 200$ and the birth-timing adjustment uses $N = 311$. In the balanced short-window panel, the raw $t = 0$ attachment gap is also 43.0 pp.

R. Why are women concentrated in non-regular employment? I

Several forces push in the same direction:

- **Regular employment is difficult to combine with care.** Long hours, overtime, transfers, and expectations of availability have historically characterised the regular-employment track (JILPT, 2019).
- **Non-regular work can offer more control over working time.** Women place particular value on flexibility in working hours when combining employment with family responsibilities (JILPT, 2019).
- **Marriage and childbirth reinforce the sorting.** Much of women's subsequent return to employment occurs through non-regular jobs, even among women who previously held regular positions (OECD, 2024).
- **Tax and social-insurance rules have historically reinforced this pattern.** Incentives facing married secondary earners encourage earnings below contribution thresholds and can encourage non-regular employment (OECD, 2024).

R. Why are women concentrated in non-regular employment? II

- **Employer demand matters too.** Firms use non-regular employment to reduce labour costs and increase workforce flexibility (OECD, 2024).

Takeaway: women facing care constraints may sort - or be pushed - into the part of the labour market that is easier to combine with family responsibilities, but that also offers weaker employment continuity.